TASK 11 – WORDPRESS HOSTING ON AWS EC2 WITH LOCAL DATABASE AND AMAZON RDS

1. Objective

To launch an AWS EC2 instance, host WordPress under a user directory, create a local database, and then connect the WordPress application to an Amazon RDS database.

The task was performed in two cases:

- **Case 1:** WordPress hosted on EC2 with a local MariaDB database.
- **Case 2:** WordPress hosted on EC2 and connected to Amazon RDS.

The WordPress website was accessed using:

```
myawshosting.net
```

Only the minimum required access was configured.

CASE 1 — WORDPRESS HOSTING WITH LOCAL DATABASE

Step 1 — Launch EC2 Instance

An EC2 instance was launched in AWS for hosting the WordPress website.

The instance was configured with a suitable Linux operating system.

The operating system was verified using:

```
cat /etc/os-release
```

Step 2 — Update the System

The system packages were updated using:

```
sudo dnf update -y
```

This ensured that the required system packages were up to date.

Step 3 — Install Apache Web Server

Apache HTTP Server was installed:

```
sudo dnf install -y httpd
```

The Apache service was enabled and started:

```
sudo systemctl enable --now httpd
```

The service status was verified using:

```
sudo systemctl status httpd
```

Step 4 — Install PHP

PHP and the required WordPress PHP packages were installed:

```
sudo dnf install -y php php-mysqlnd php-gd php-mbstring php-xml php-json
php-curl
```

Apache was restarted:

```
sudo systemctl restart httpd
```

Step 5 — Download WordPress

The WordPress package was downloaded:

```
cd /tmp
wget https://wordpress.org/latest.tar.gz
```

The downloaded archive was extracted:

```
tar -xzf latest.tar.gz
```

Step 6 — Host WordPress Under User Directory

A directory was created under the EC2 user's home directory:

```
mkdir -p /home/ec2-user/wordpress
```

The WordPress files were copied into the directory:

```
cp -r /tmp/wordpress/* /home/ec2-user/wordpress/
```

The WordPress application was therefore hosted under:

```
/home/ec2-user/wordpress
```

Step 7 — Configure Apache

Apache was configured to serve the WordPress files from the user directory.

The WordPress directory was configured as the web document root.

The required directory permissions were also configured so that Apache could access the WordPress files.

Step 8 — Install Local MariaDB

MariaDB was installed on the EC2 instance:

```
sudo dnf install -y mariadb105-server
```

The MariaDB service was enabled and started:

```
sudo systemctl enable --now mariadb
```

Step 9 — Create WordPress Database

MariaDB was accessed using:

```
sudo mysql
```

A database named `wordpress` was created:

```
CREATE DATABASE wordpress;
```

A database user was created:

```
CREATE USER 'wpuser'@'localhost' IDENTIFIED BY 'Wp@ss12345';
```

Required privileges were granted:

```
GRANT ALL PRIVILEGES ON wordpress.* TO 'wpuser'@'localhost';
```

The privileges were applied:

```
FLUSH PRIVILEGES;
```

Step 10 — Configure WordPress Database

The WordPress configuration file was created:

```
cd /home/ec2-user/wordpress
sudo cp wp-config-sample.php wp-config.php
```

The configuration file was edited:

```
sudo vim wp-config.php
```

The database details were configured as:

```
define( 'DB_NAME', 'wordpress' );
define( 'DB_USER', 'wpuser' );
define( 'DB_PASSWORD', 'Wp@ss12345' );
define( 'DB_HOST', 'localhost' );
```

Step 11 — Set Permissions

Ownership was assigned to Apache:

```
sudo chown -R apache:apache /home/ec2-user/wordpress
```

The required permissions were configured:

```
sudo chmod -R 755 /home/ec2-user/wordpress
```

Step 7 — Configure SELinux for Database Connectivity

```
sudo setsebool -P httpd_can_network_connect_db 1
```

Then:

```
getsebool httpd_can_network_connect_db
```

Step 12 — Verify Local Database WordPress

The WordPress website was opened using the configured domain:

```
myawshosting.net
```

The WordPress installation page was displayed successfully.

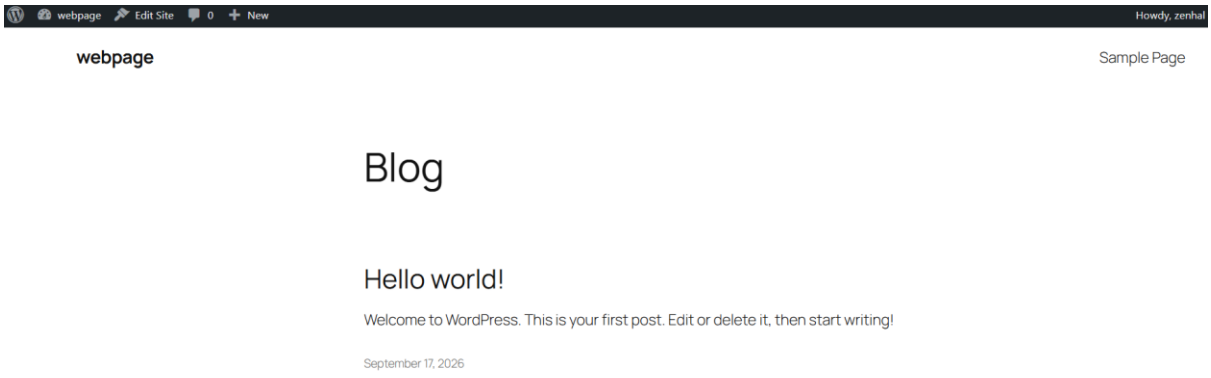
The WordPress installation was completed by providing the required site and administrator details.

CASE 1 VERIFICATION

The following operations were successfully completed:

- EC2 instance launched.
- Apache web server installed and configured.
- PHP installed.
- WordPress downloaded and hosted under the user directory.

- Local MariaDB installed.
- WordPress database created.
- Database user and privileges configured.
- WordPress connected to the local database.
- WordPress webpage accessed successfully.



CASE 2 — WORDPRESS WITH AMAZON RDS

Step 1 — Create Amazon RDS Instance

An Amazon RDS MySQL database instance was created for storing the WordPress database.

The RDS instance was configured with the required database settings.

Step 2 — Configure RDS Security Group

The RDS Security Group was configured to allow database access only from the required EC2 Web Server.

The MySQL port used was:

3306

The source was restricted to the required Web Server Security Group.

No unnecessary public database access was provided.

Step 3 — Create WordPress Database in RDS

A WordPress database was created in the RDS MySQL instance.

A dedicated database user was configured with the required privileges for the WordPress database.

Step 4 — Test RDS Connectivity

The connection from the EC2 Web Server to the RDS database was tested using the MySQL client:

```
mysql -h <RDS-ENDPOINT> -u <RDS-USERNAME> -p
```

The connection was successfully established.

Step 5 — Configure WordPress for RDS

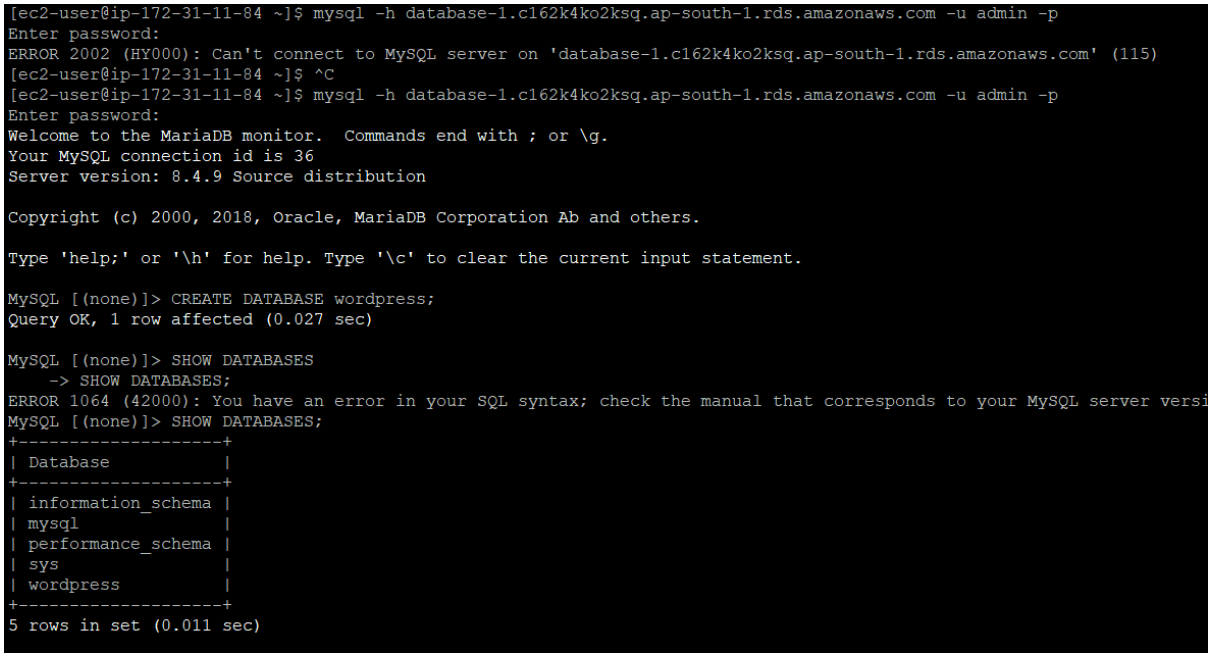
The WordPress configuration file was modified:

```
sudo vim /home/ec2-user/wordpress/wp-config.php
```

The database configuration was changed from the local database to the RDS database:

```
define( 'DB_NAME', 'wordpress' );
define( 'DB_USER', '<RDS-USERNAME>' );
define( 'DB_PASSWORD', '<RDS-PASSWORD>' );
define( 'DB_HOST', '<RDS-ENDPOINT>' );
```

The RDS endpoint was used as the database host instead of `localhost`.



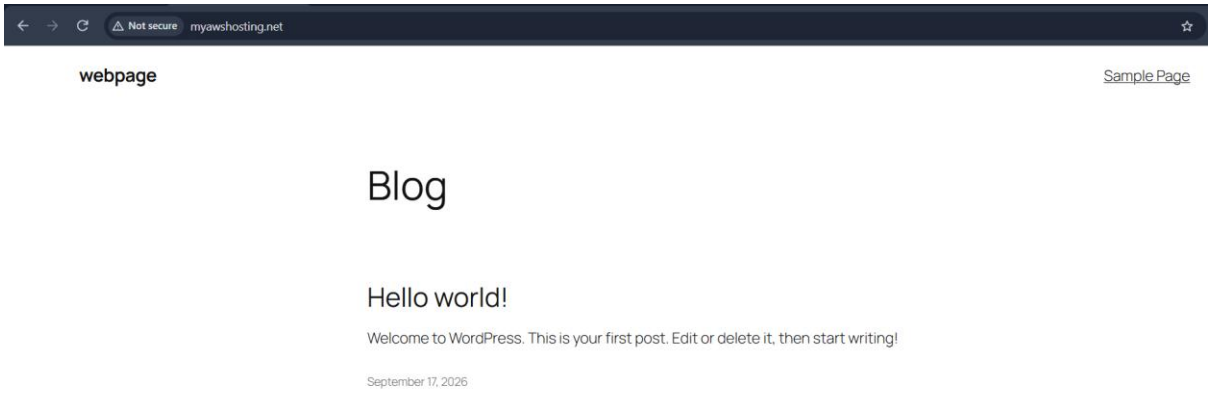
Step 6 — Access WordPress Using Domain

The WordPress website was accessed using:

`myawshosting.net`

The WordPress webpage was displayed successfully.

The website was therefore running on the EC2 Web Server while the database was hosted separately on Amazon RDS.



RDS VERIFICATION

The following operations were successfully verified:

- Amazon RDS MySQL instance created.
- RDS Security Group configured.
- MySQL port 3306 allowed only from the required source.
- WordPress database configured in RDS.
- EC2 Web Server connected successfully to RDS.
- WordPress configuration updated with RDS endpoint.
- WordPress webpage accessed successfully through `myawshosting.net`.
- Minimum required access was maintained.

RESULT

- WordPress was successfully hosted on an AWS EC2 instance under a user directory.
- The WordPress application was first configured with a local MariaDB database and then connected to an Amazon RDS MySQL database.
- The website was successfully accessed using:
- `myawshosting.net`

- Only the minimum required access was granted for the database connection

CONCLUSION

The practical successfully demonstrated WordPress hosting using AWS EC2 with both a local database and Amazon RDS.

The first case demonstrated WordPress hosting with a local MariaDB database. The second case demonstrated separation of the web application and database by hosting WordPress on EC2 and the database on Amazon RDS.

The RDS Security Group was configured to provide only the required database access, following the principle of **minimum access**.